ANESTHESIOLOGY

Gastric Pressure Monitoring Unveils Abnormal Patient–Ventilator Interaction Related to Active Expiration: A Retrospective Observational Study

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ABSTRACT

Background: Patient–ventilator dyssynchrony is frequently observed during assisted mechanical ventilation. However, the effects of expiratory muscle contraction on patient–ventilator interaction are underexplored. The authors hypothesized that active expiration would affect patient–ventilator interaction and they tested their hypothesis in a mixed cohort of invasively ventilated patients with spontaneous breathing activity.

Methods: This is a retrospective observational study involving patients on assisted mechanical ventilation who had their esophageal pressure (P_{eso}) and gastric pressure monitored for clinical purposes. Active expiration was defined as gastric pressure rise (ΔPgas) greater than or equal to 1.0 cm H_2O during expiratory flow without a corresponding change in diaphragmatic pressure. Waveforms of P_{eso}, gastric pressure, diaphragmatic pressure, flow, and airway pressure (P_{aw}) were analyzed to identify and characterize abnormal patient–ventilator interaction.

Results: A total of 76 patients were identified with P_{eso} and gastric pressure recordings, of whom 58 demonstrated active expiration with a median ΔPgas of 3.4 cm H_2O (interquartile range = 2.4 to 5.3) observed in this subgroup. Among these 58 patients, 23 presented the following events associated with expiratory muscle activity: (1) distortions in P_{aw} and flow that resembled ineffective efforts, (2) distortions similar to autotriggering, (3) multiple triggering, (4) prolonged ventilatory cycles with biphasic inspiratory flow, with a median percentage (interquartile range) increase in mechanical inflation time and tidal volume of 54% (44 to 70%) and 25% (8 to 35%), respectively and (5) breathing exclusively by expiratory muscle relaxation. Gastric pressure monitoring was required to identify the association of active expiration with these events. Respiratory drive, assessed by the rate of inspiratory P_{eso} decrease, was significantly higher in patients with active expiration (median [interquartile range] dP_{eso}/dt: 12.7 [9.0 to 18.5] *vs* 9.2 [6.8 to 14.2] cmH_2O/sec; $P < 0.05$).

Conclusions: Active expiration can impair patient–ventilator interaction in critically ill patients. Without documenting gastric pressure, abnormal patient–ventilator interaction associated with expiratory muscle contraction may be mistakenly attributed to a mismatch between the patient's inspiratory effort and mechanical inflation. This misinterpretation could potentially influence decisions regarding clinical management.

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EDITOR'S PERSPECTIVE

What We Already Know about This Topic

- Patient ventilator dyssynchrony is common during prolonged mechanical ventilation in critically ill patients and can be associated with adverse outcome and prolonged weaning
- This can involve either inspiration or expiration, although less is known regarding the effects of expiratory musculature on it
- Recognition requires careful inspection of ventilator waveforms along with esophageal and gastric pressures

What This Article Tells Us That Is New

- In a retrospective analysis of a cohort of ventilated critically ill patients, waveforms and pressure tracings were evaluated by the investigators
- Based on elevations of gastric pressure during expiratory flow, they categorized five unique groups based on expiratory muscle activity
- Respiratory drive was significantly higher in subjects with active expiration
- This approach holds promise to better clarify dyssynchrony with regard to which respiratory phase is predominant

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Harmonious patient–ventilator interaction represents one of the most complex issues during assisted mechanical ventilation.[1] Patient–ventilator dyssynchrony occurs in approximately 40% of respiratory cycles, with an incidence varying among studies, depending on the population studied, ventilator settings, detection method, and monitoring duration.[2–4] There is evidence that dyssynchronies are linked to the duration of mechanical ventilation, length of intensive care unit (ICU) and hospital stay, and possibly, mortality.[4–6] Over the years, different patterns of abnormal patient–ventilator interaction have emerged, with reverse triggering being the most recent.[7] They are mostly detected by careful observation of airway pressure and flow waveforms, and, in some cases through inspiratory muscle activity monitoring, assuming that abnormal interaction primarily refers to a mismatch between the patient's inspiratory effort and the provision of ventilatory support. The expiratory muscles are rarely, if ever, monitored and their contribution to patient–ventilator interaction is largely neglected.

Expiratory muscles primarily include the abdominal muscles and the intercostal internal muscles. Their contraction facilitates exhalation by increasing abdominal pressure and moving the diaphragm upwards into the thoracic cavity. The resulting increase in pleural pressure decreases lung volume.[8] Generally, these muscles are inactive during normal resting breathing, but they are recruited when there is an imbalance between the load imposed on inspiratory muscles (*e.g.*, exercise, impaired respiratory system mechanics, increased partial pressure of carbon dioxide, and others) and their capacity to respond to this load (*e.g.*, respiratory muscle weakness).[8–10] High respiratory load and impaired contracting ability of inspiratory muscles are commonly observed in critically ill patients. Thus, expiratory muscle activation is not uncommon in spontaneously breathing, critically ill patients and has been associated with positive and negative effects, such as improved diaphragm neuromechanical coupling during loaded breathing but also a higher likelihood of weaning failure.[11–15] More recently, Jonkman *et al.*,[16] reported ventilator triggering by expiratory muscles relaxation in a critically ill patient monitored with esophageal pressure (P_{eso}) and gastric pressure. Apart from this case report, there are no studies investigating the role of expiratory muscles in patient–ventilator interaction, likely due to the complexity of expiratory muscle contraction monitoring, which requires the recording of gastric pressure.

We hypothesized that under assisted mechanical ventilation, active expiration would impair patient–ventilator interaction. We investigated our hypothesis in a cohort of mixed adult ICU patients in whom P_{eso} and diaphragmatic pressure were monitored for clinical purposes. We explored (1) whether active expiration was associated with distortions of ventilator waveforms that could be interpreted as dyssynchronies and (2) whether these would be misclassified and misinterpreted without gastric pressure documentation.

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Materials and Methods

This is a retrospective observational study and includes a mixed population of invasively ventilated patients admitted to the medico-surgical adult ICU of the University Hospital of Heraklion (Iraklio, Greece) between 2018 and 2022. Eligible for inclusion and analysis were all patients with spontaneous breathing activity, without neuromuscular blocking agent administration, under control-assisted or fully assisted invasive mechanical ventilation with available recordings of minute ventilation ($\dot{V}_E$), airway pressure (P_{aw}), P_{eso}, and gastric pressure. The indications for measuring P_{eso} were either the use of lung protective ventilation in patients with severe acute respiratory distress syndrome (ARDS), or optimization of patient–ventilator interaction and titration of assist. To monitor P_{eso}, we regularly use a nasogastric catheter with esophageal and gastric balloons (NutriVent TM, Nutrivent, Italy). This catheter enables both feeding and P_{eso} and gastric pressure measurements. By subtracting P_{eso} from gastric pressure, diaphragmatic pressure is also available, further aiding inspiratory muscle assessment.[17] The proper position and filling of the esophageal balloon were verified using standard tests.[17] The use of anonymized data was approved by the hospital Ethics Committee. Because this was an observational study that did not involve therapeutic or clinical intervention, informed consent was not required.

Measurements and Data Analysis

Included patients were ventilated with Evita-XL (Drager, Germany) or Puritan-Bennett 840 or 980 ventilator (Medtronic, PLC, Ireland). Airflow and pressures were measured using a heated pneumotachograph (Hans-Rudolf 3700, USA), and differential pressure transducers (Micro-Switch 140PC, Honeywell, Canada). Tidal volume (V_T) was obtained by integrating the flow signal. All signals were sampled and stored at 200 Hz (Windaq Instruments, USA) and analyzed offline. Some of these recordings were included in the analysis of previously published studies. In these studies, the contribution of active expiration to patient–ventilator interaction was not addressed.[18–20]

Four investigators with extensive experience in the analysis and interpretation of flow, P_{aw}, P_{eso}, and gastric pressure waveforms (E.A., K.V., E.K., D.G.), retrospectively reviewed all recordings from eligible patients. Recordings or parts of them with artifacts related to signal measurement, catheter malposition, or esophageal peristalsis were discarded as invalid. The analysis proceeded as follows.

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Step 1. At first, it was assessed whether active expiration was present in valid recordings. The latter is defined in the literature as an increase in gastric pressure (ΔPgas) greater than or equal to 1.0 cmH_2O from its lowest value after the end of inspiratory flow to its peak value during expiration (ΔPgas) (Supplemental Figure E1, https://links.lww.com/ALN/D570).[20,21] Cardiogenic oscillations may induce changes in ventilator pressures up to 2 to 3 cmH_2O and have a frequency between 0.8 to 4 Hz.[22,23] Cardiac oscillations were distinguished from active expiration, based on the characteristics of gastric pressure increase: gastric pressure distortions induced by cardiac impulses should be repetitive, short-term, align with similar periodic fluctuations in P_{eso} and P_{aw}, and occur throughout the respiratory cycle.

Step 2. Flow, P_{aw}, and waveforms of recordings with active expiration were meticulously examined, during their entire recording period for deviations suggestive of poor patient–ventilator interaction.[1] When such deviations occurred, an inspection of P_{eso}, diaphragmatic pressure, and gastric pressure waveforms was carried out to identify whether they were related to gastric pressure and not to diaphragmatic activity. It was also evaluated whether the morphology of the expiratory flow-time curve indicated the presence of active expiration. Only recordings where all four experts unanimously agreed on the identification and classification of abnormal patient–ventilator interaction events were included in the analysis.

Finally, we measured drive in all eligible patients and recordings. The respiratory drive was assessed in two ways, the rate of diaphragmatic pressure increase (dPdi/dt) and the rate of P_{eso} decrease (dP_{eso}/dt) during inspiration, and a value exceeding 10 cmH_2O/sec was regarded as high based on previous observations.[19,24,25] We also collected available data on factors related to active expiration, such as maximal inspiratory muscle pressure, the magnitude of inspiratory effort, the use of opioids, the level of positive end-expiratory pressure (PEEP), and ventilator assistance. Measured parameters are detailed in the Supplemental Digital Content (https://links.lww.com/ALN/D570).

Statistical Analysis

Patient demographics, clinical features, and ventilation characteristics are presented as medians and interquartile ranges. Categorical variables are presented as percentages. We did not estimate sample size *a priori* for this study. A Mann-Whitney U test was conducted to compare differences in fentanyl dose, PEEP level, inspiratory diaphragmatic pressure swing, P_{eso} swing, V_T, respiratory rate (RR), dPdi/dt, dP_{eso}/dt, maximal inspiratory pressure, and P0.1 between patients with active and passive expiration, excluding cases test by test. A Spearman's rank correlation analysis was run to assess the relationship between dPdi/dt and ΔPgas and dP_{eso}/dt and ΔPgas. In the 35 patients who had recordings under different ventilator settings, linear mixed-effect model analysis was performed with ΔPgas as the dependent variable, fentanyl administration, presence of any level of assist (pressure support ventilation or proportional assist ventilation with load adjustable gain factors *vs.* continuous positive airway pressure), and high respiratory drive as fixed effects, and each subject as random effect. All statistical tests were two-tailed, and $P < 0.05$ was considered statistically significant. IBM SPSS Statistics for Windows, version 23 (USA), was used for the analysis.

Results

Baseline Patient Characteristics and Frequency of Expiratory Muscle Contractions

Recordings from 83 patients with active breathing and recordings of P_{aw}, $\dot{V}_E$, P_{eso}, and gastric pressure were explored. Seven patients were excluded due to unreliable waveforms. Hence, 76 patients were analyzed (fig. 1). Table 1 shows patients' and ventilator parameters. Records with more than one ventilation mode or multiple levels of support were available in 35 patients. By considering each record as a separate condition, 123 individual conditions, with a duration of 26.4 min (18.8 min to 49.8 min) were identified and analyzed. Among the 123 conditions, 66 (54%) were in pressure support ventilation, 28 (23%) in proportional assist ventilation, 25 (20%) in continuous positive airway pressure and 4 (3%) in pressure assist control. Active expiration was documented in 58 of the 76 patients (76%) and 88 of the 123 conditions (72%).

Abnormal Patient–Ventilator Interaction Associated With Expiratory Muscle Activity

Expiratory muscle contraction impaired patient–ventilator synchrony in 23 out of 58 patients with active expiration. In 6 recordings there was uncertainty among investigators about the type of impaired patient–ventilator interaction or whether this was related to gastric pressure and they were excluded. The following events of abnormal patient-ventilator interaction associated with active expiration were noted (table 2, fig. 1):

1. Events that could be interpreted as ineffective efforts (fig. 2). They occurred throughout the recording period and were apparent in continuous positive airway pressure, pressure support ventilation, and proportional assist ventilation with load adjustable gain factors.
2. Events resembling autotriggering not caused by cardiac oscillations, liquids in the circuit, or tracheobronchial secretions but by fluctuations of expiratory muscle activity. The resulting V_T and increase in minute ventilation depended on the frequency and magnitude of gastric pressure variation, the mechanical properties of the respiratory system, and the level of ventilatory assist (fig. 3). These events occurred sporadically, once every 2

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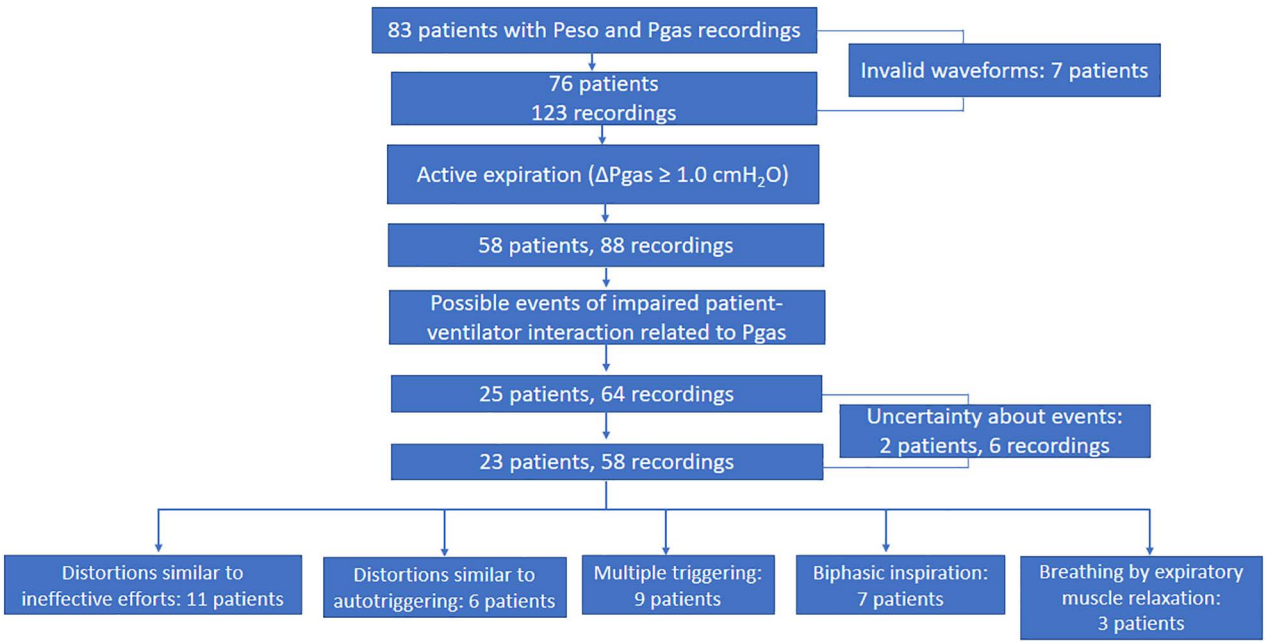

Fig. 1. Flow diagram of the results. P_{eso}, esophageal pressure; Pgas, gastric pressure.

to 5 min in three patients, or more regularly, several times per minute, in the rest. There were no autotriggering events during proportional assist ventilation with load adjustable gain factors ventilation.

3. Multiple triggering (fig. 4A-C). In seven out of nine patients, multiple triggering occurred throughout the recording period. Multiple triggering was never observed during ventilation with proportional assist ventilation with load adjustable gain factors. In three patients receiving assisted controlled mechanical ventilation, multiple trigger had a repetitive pattern that gave the impression of entrainment. Nevertheless, careful observation revealed that the mechanism of triggering varied between breaths (fig. 5),[7] thus excluding entrainment as the cause of multiple triggering.
4. Biphasic inspiration cycles (see breaths 1 and 4 in fig. 6) with a median % (interquartlie range) increase in mechanical inflation time and V_T of 54% (44 to 70%) and 25% (8 to 35%) compared to the other cycles, respectively. Without gastric pressure monitoring, it could be assumed that the mechanism for biphasic inspiration was a delayed cycling-off (note the P_{aw} peak at the end of inflation suggesting a late cycling-off) or a biphasic increase in diaphragmatic pressure.
5. Breathing exclusively by relaxation of the expiratory muscles. In two out of three patients this occurred during proportional assist ventilation with load adjustable gain factors ventilation. Since gastric pressure increased by 1.5 to 2.0 cmH_2O during occluded breaths, the plateau P_{aw} did not correspond to passive conditions and the calculated driving pressure was overestimated (Supplemental Figure E2, https://links.lww.com/ALN/D570).[20]

Table 1. Characteristics of the Population on the Study Day

Patient characteristics, N = 76	Age, yr	71 (63–77)
	Male, n (%)	47 (62%)
	BMI, kg/m²	30 (24–35)
	APACHE-II score	17 (14–21)
	RASS	–3 (–4 to –1)
Cause for invasive MV, N (%)	Hypoxemic RF	35 (46%)
	Hypercapnic RF	5 (7%)
	Sepsis	7 (9%)
	Cardiorespiratory arrest	6 (8%)
	neurologic emergency	11 (14%)
	Surgery/trauma	9 (12%)
	Others	3 (4%)
Ventilation characteristics	Pa_{O_2}/Fi_{O_2}	235 (200–304)
	PEEP, cmH_2O	8 (7–9)
	Level of assist	
	PSV, cmH_2O	7 (5–8)
	PAV+, % assist	40% (35–50%)
	Days on MV before study	9 (5–12)
Outcomes	ICU LOS, days	21 (11–35)
	MV duration, days	14 (7–21)
	ICU survival	58 (66%)

Values are median (1st, 3rd quartile) or number (% of total).

APACHEII, Acute Physiology and Chronic Health Evaluation Score; BMI, body mass index; ICU, intensive care unit; LOS, length of stay; MV, mechanical ventilation; PAV+, proportional assist ventilation; PEEP, positive end expiratory pressure; PSV, pressure support ventilation; RASS, Ritchmond Agitation and Sedation Scale; RF, respiratory failure.

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Table 2. Types of Events

Type of Event	Appearance	Mechanism	Patients
IE resembling events	Decreases in expiratory flow without inspiratory muscle contraction	Sudden short-term decrease in Pgas during active expiration	11
Autotriggering resembling events	Ventilator cycles triggered without inspiratory effort	Short-term decrease in Pgas during active expiration that triggered a cycle	6
Multiple cycles	≥ 2 ventilator cycles without complete expiration between them	Decrease in Pgas triggered a cycle during partially or fully assisted ventilation	9
Biphasic inspiration	Two-phase increase in inspiratory flow, longer inflation, maybe higher V_T	Pgas decrease triggered the first inspiratory flow increase, subsequent diaphragmatic contraction produced the second increase	7
Breathing by expiratory muscle relaxation	Regularly triggered cycles No diaphragmatic contraction	Triggering by expiratory muscle relaxation	3

IE, ineffective efforts; Pgas, gastric pressure; V_T, tidal volume.

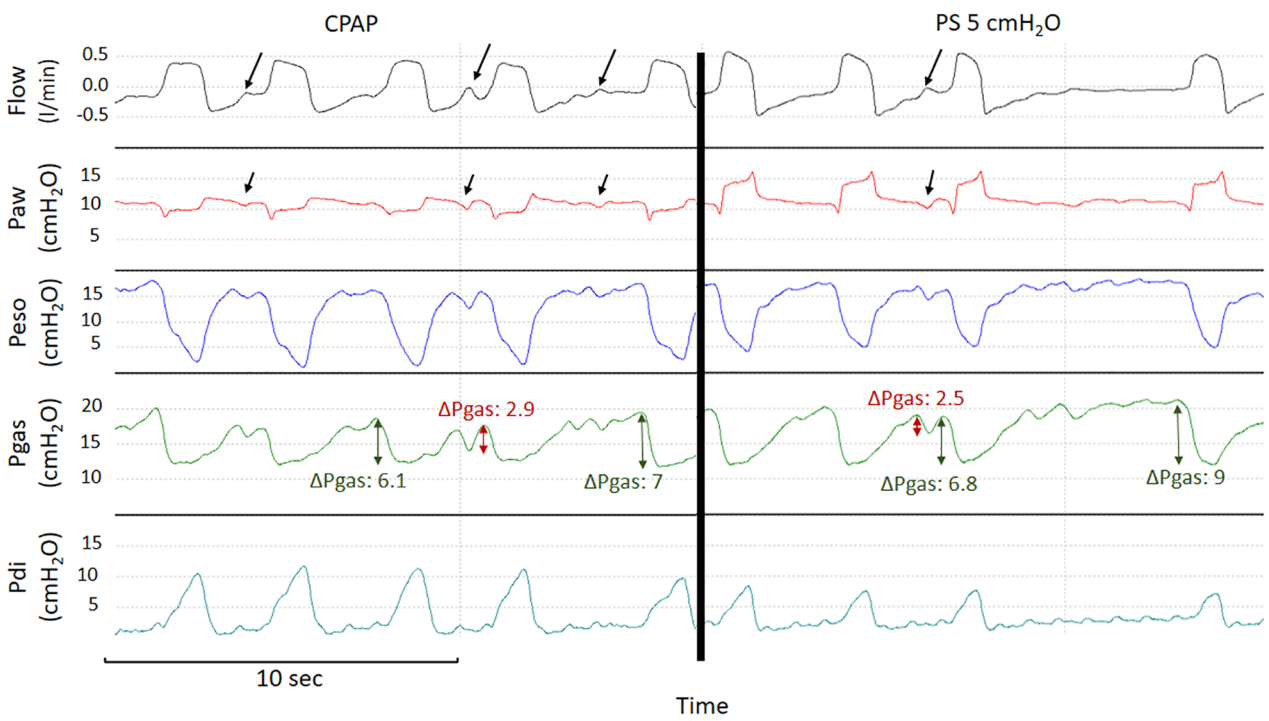

Fig. 2. Flow, airway pressure (P_{aw}), esophageal pressure (P_{eso}), gastric pressure (Pgas), and transdiaphragmatic pressure (Pdi) in a patient ventilated on continuous positive airway pressure (CPAP) with positive end-expiratory pressure of 11 cm H_2O (*left*) and with Pressure Support Ventilation (PSV) with a level of support of 5 cmH_2O and PEEP of 11 cmH_2O (*right*). Notice the abrupt decreases in expiratory flow associated with decreases in P_{aw} (*black arrows*) that resemble ineffective efforts. There is no diaphragmatic contraction and the distortions in expiratory flow and P_{aw} are caused by Pgas fluctuations during expiratory muscle contraction. Contraction of expiratory muscles causes an increase in Pgas (ΔPgas) between 6.1 to 9.0 cmH_2O (*double green arrows*). The magnitude of Pgas fluctuations was 2.5 to 2.9 cmH_2O (*double red arrows*).

Use of Ventilator Waveform Monitoring to Diagnose Active Expiration

Passive expiration is characterized by an exponential decay of expiratory flow (with a triangular shape) toward zero flow (in the absence of dynamic hyperinflation) before inspiration. Active expiration, however, distorts the passive expiratory flow shape and prevents the expiratory flow from reaching zero at the end of expiration. Observing the expiratory flow-time shape is one way to detect expiratory muscle activity.[1,26] In 32% of the recordings with active expiration, the expiratory flow-time shape was similar to passive expiration and the expiratory flow decreased to zero. Thus, the ventilator waveforms gave no evidence of active expiration (figs. 3 and 4, B and C).

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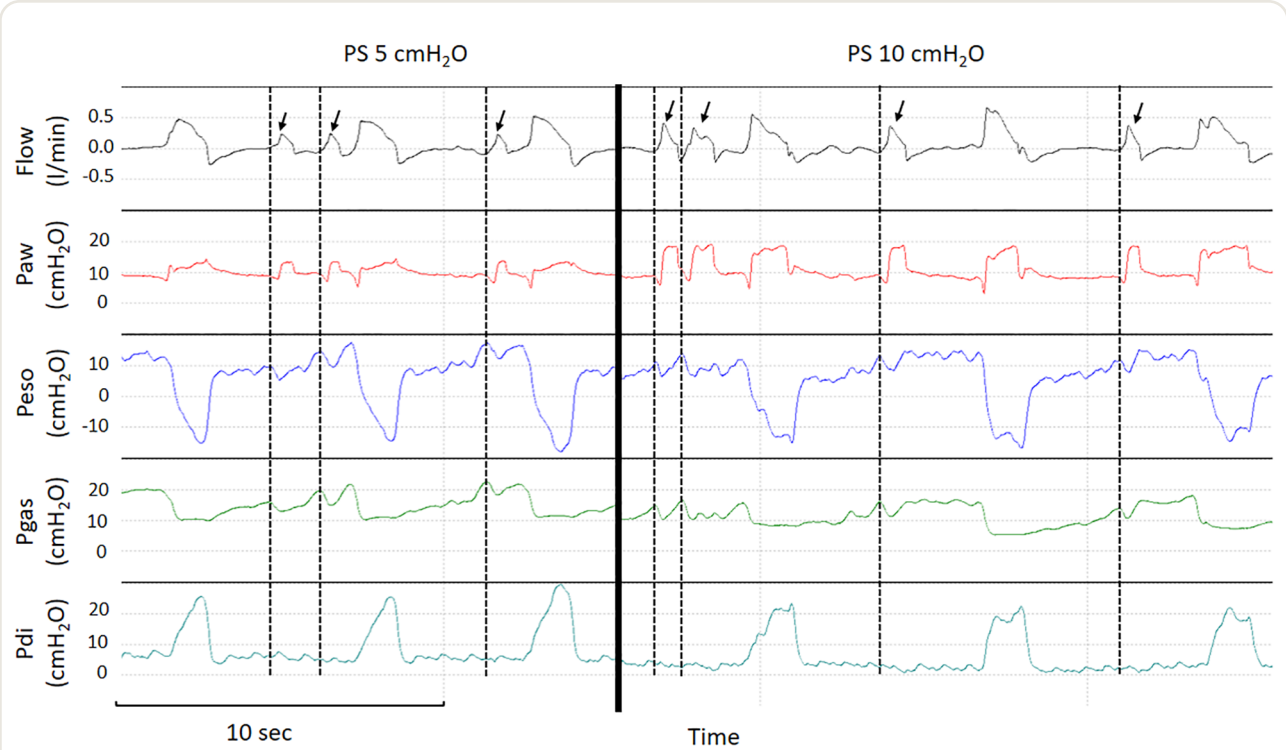

Fig. 3. Triggering by gastric pressure (Pgas) fluctuations in a patient ventilated with pressure support (PS) 5 (*left*) and 10 cmH_2O (*right*). Notice multiple ventilator triggering (*black arrows*) caused by Pgas fluctuations of 3 to 6 cmH_2O (*grey dotted lines* denote the beginning of partial relaxation). Tidal volume of ventilator breaths triggered by inspiratory muscle efforts was 390 to 400 ml during PS of 5 cmH_2O and 460 to 470 ml during PS of 10 cmH_2O. Tidal volume of cycles triggered by Pgas fluctuations was 60 to 90 ml and 130 to 140 ml during PS 5 and PS 10 cmH_2O, respectively. ΔPgas was 9.5 to 10 cmH_2O in the *left* and 9 to 11 cmH_2O in the *right*. The absence of expiratory flow during mechanical expiration can be misinterpreted as passive expiration. Pdi, diaphragmatic pressure; P_{eso}, esophageal pressure

The median (interquartile range) duration of recordings with active expiration was 27.2 min (17.9 to 48.3 min). When active expiration was present, it persisted throughout the recording. Expiratory muscle contraction constituted 59% (51 to 64%) of the respiratory cycle with a minimum and maximum value of 32% and 93%, respectively. In two patients, ventilated with assist control mechanical ventilation, active expiration lasted 4 s and 14 s (fig. 7). Regular ventilator triggering occurred secondary to gastric pressure fluctuations over an extremely intense and prolonged active expiration period.

Effects of Respiratory Drive and Effort, Ventilator Settings, and Medication on Active Expiration

Table 3 shows measurements of inspiratory and expiratory effort, ventilation, and indices of respiratory drive.

There was no significant effect of fentanyl administration (P = 0.32), presence of any ventilator assist (pressure delivery *vs.* continuous positive airway pressure) (P = 0.7), and high respiratory drive (P = 0.21). Moreover, fentanyl dose, PEEP level, level of assist, diaphragmatic pressure swing, V_T, RR, maximal inspiratory muscle pressure, and P0.1 did not significantly differ between conditions of active and those with passive expiration (table 4). However, respiratory drive, as estimated by dP_{eso}/dt, but not dPdi/dt, and ΔP_{eso} was significantly higher in the active expiration group. Of note, there were missing values on these parameters (refer to Supplemental Digital Content, https://links.lww.com/ALN/D570). The characteristics and respiratory drive of patients with events of abnormal patient–ventilator interaction as a result of active expiration is shown in Supplemental Tables E1-3 (https://links.lww.com/ALN/D570). There was a significant (P = 0.006), albeit weak (r = 0.288) positive correlation between dP_{eso}/dt and ΔPgas, but no significant correlation between dPdi/dt and ΔPgas (r = 0.08, P = 0.47).

Discussion

This retrospective observational study examined how expiratory muscle contraction affected patient–ventilator interaction in a large group of ICU patients receiving assisted mechanical ventilation. Three-quarters of patients had active expiration, defined as ΔPgas ≥ 1.0 cmH_2O. These patients had higher respiratory drive, and in more than one-third of them, active expiration was associated with significant events of abnormal patient–ventilator interaction, such as multiple cycles and autotriggering. Furthermore, gastric

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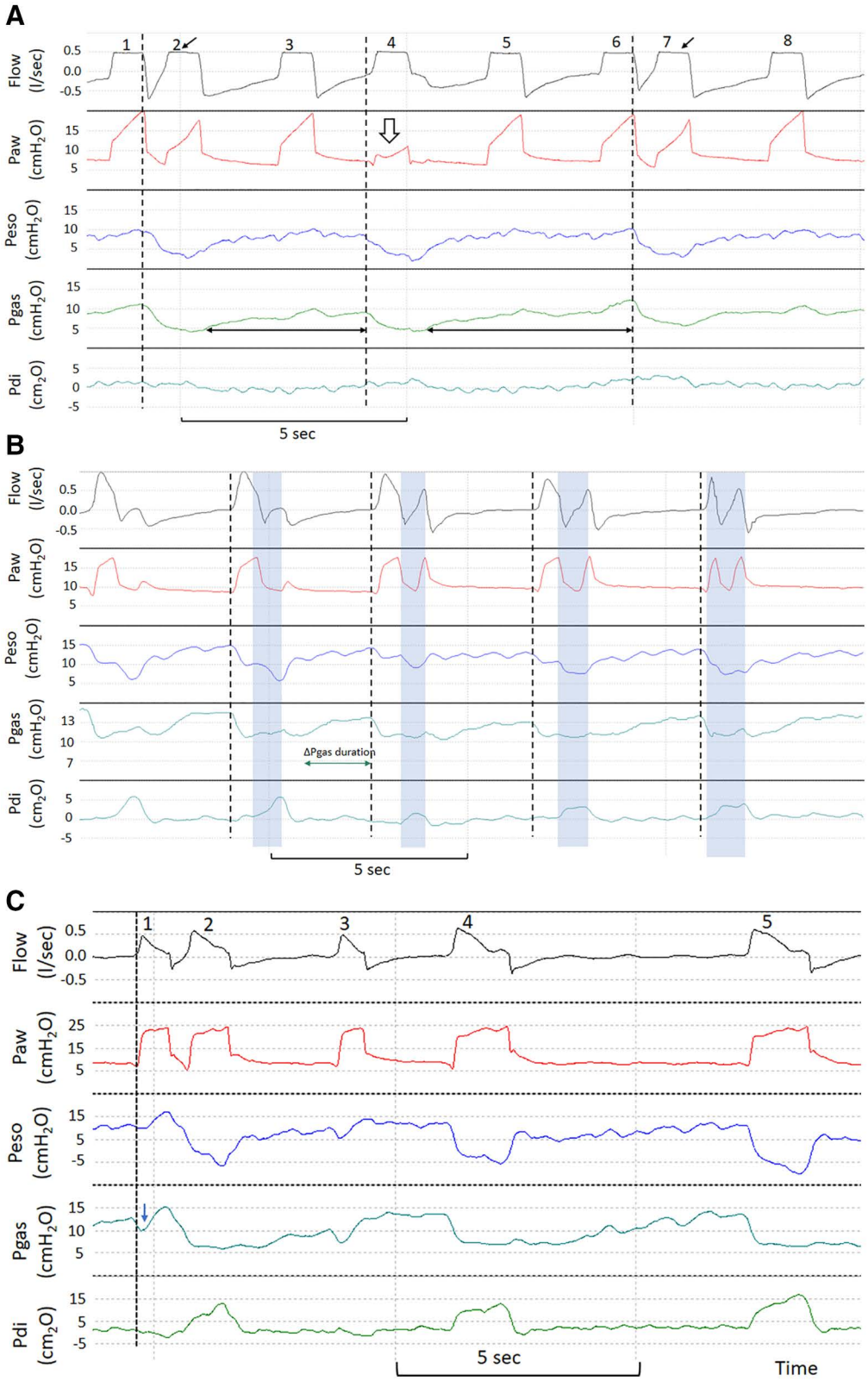

Fig. 4. Flow, airway pressure (P_{aw}), esophageal pressure (P_{eso}), gastric pressure (Pgas), and transdiaphragmatic pressure (Pdi) in three different patients ventilated with assist volume control (*A*) and pressure support (*B* and *C*). (*A*) During assist volume control, flow, P_{aw} and *(Continued)*

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Fig. 4. *(Continued)* P_{eso} waveforms in breaths 2,4,7 could be misinterpreted as signifying inspiratory muscle activity causing either stacked breaths (*black arrows*) or significant decrease in P_{aw} (*open arrow*). However, Pdi remained rather constant throughout the recording, suggesting no or minimal diaphragmatic activity. Inspection of Pgas shows significant expiratory muscle activity which is maintained even during the mechanical inflation time (breaths 3, 5 and 8). Relaxation of expiratory muscles either triggered the ventilator early in expiratory phase (breaths 2 and 7) or considerably decreased P_{aw} (*open arrow*, breath 4). Note that although breath-2 was added to the first one, P_{aw} early during the second mechanical inflation is less than during the first mechanical inflation due to the decrease in alveolar pressure caused by expiratory muscle relaxation. *Double edges horizontal arrows* indicate the duration of expiratory muscle activity. (*B*) During Pressure Support ventilation, the ventilator was triggered first by expiratory muscle relaxation and then by inspiratory efforts, leading to double triggering. *Vertical dotted lines* indicate the relaxation of expiratory muscles. *Blue shaded area* indicates the duration of diaphragmatic contraction. (*C*) During Pressure Support ventilation, Pgas fluctuation (*blue arrow*) triggered the first breath and the second breath was triggered by the inspiratory effort along with complete relaxation of expiratory muscles. The third breath was only triggered by Pgas fluctuation.

pressure fluctuations caused P_{aw} and flow distortions that resembled ineffective efforts, even during continuous positive airway pressure. Analyzing the ventilator waveforms alone would not reveal the mechanism of these events. Even advanced monitoring with esophageal pressure was not sufficient to identify expiratory muscle activation as the mechanism of P_{aw} and flow changes.

Methodologic Aspects

We calculated ΔPgas as the change in gastric pressure from the lowest to the highest value during the expiratory flow. This value does not take into account the expiratory muscles of the rib cage and may be influenced by the post-inspiratory activity of the diaphragm, but it is the index that best correlates with abdominal electromyographic activity.[21,27] Additionally,

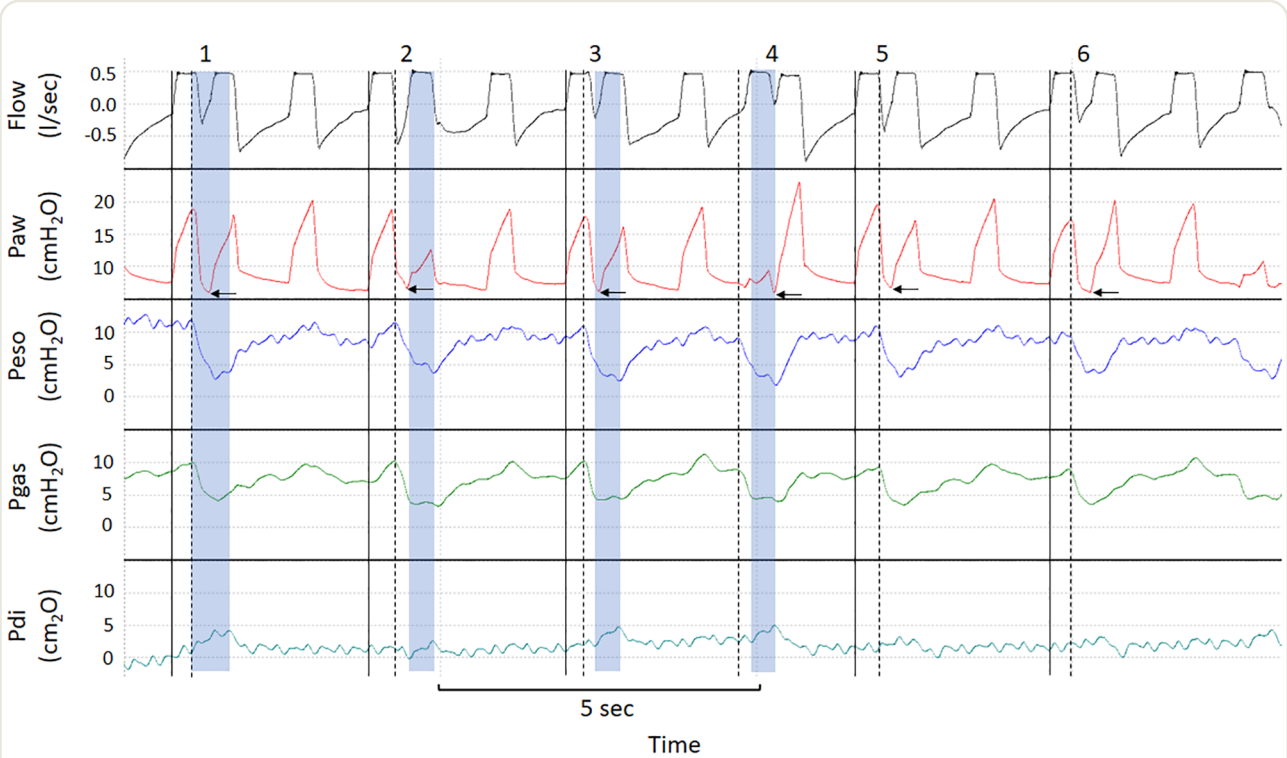

Fig. 5. Flow, airway pressure (P_{aw}), esophageal pressure (P_{eso}), gastric pressure (Pgas), and transdiaphragmatic pressure (Pdi) in a patient ventilated with assist volume control ventilation. Although peak Pdi is low (1 to 2 cmH_2O, end of *blue shaded areas*) or zero, six double triggering breath complexes, caused by different mechanisms, were observed. In these complexes (1 to 6) *vertical solid lines* show the time of ventilator-triggered breaths and *dashed lines* the start of expiratory muscles relaxation. The first breath was triggered by the ventilator in complexes no. 1, 2, 3, 5 and 6, and by relaxation of expiratory muscles in complex no. 4. The second breath was triggered (*black horizontal arrow* in P_{aw} waveform shows the triggering point) by a combined action of diaphragmatic contraction and expiratory muscles relaxation in complexes no. 1, 3 and 4 (Pdi increased and Pgas decreased before the point of triggering). The second breath in complex no. 2 was triggered only by relaxation of expiratory muscles, since Pdi starts to increase after the point of triggering. Similarly, the second breath in complexes 5 and 6 was triggered by relaxation of expiratory muscles, since no diaphragmatic contraction was observed.

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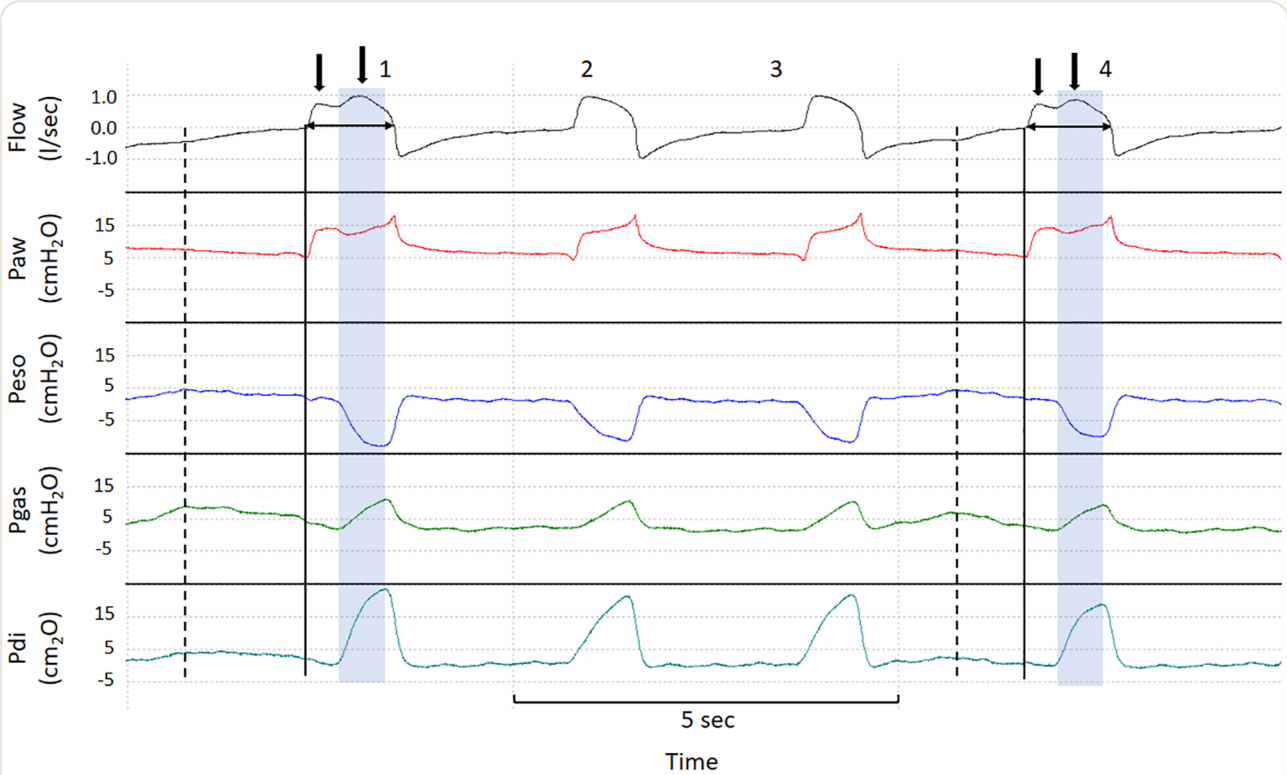

Fig. 6. Flow, airway pressure (P_{aw}), esophageal pressure (P_{eso}), gastric pressure (Pgas), and transdiaphragmatic pressure (Pdi) in a patient ventilated with pressure support. Notice that in some breaths (1 and 4) the inspiratory flow-time waveform exhibits a biphasic pattern (*black arrows*). Pgas waveform clearly indicates that the first peak was due to ventilator triggering (*vertical solid lines*) caused by the slow relaxation of expiratory muscles starting at mid-range of mechanical deflation (*vertical dashed lines*), while the second peak was due to diaphragmatic contraction (*blue areas*) occurring after the relaxation of expiratory muscles. This phenomenon did not occur in every breath since expiratory muscle contraction had a sporadic pattern (2 out of 4 breaths). Compared to other breaths, the two breaths with the biphasic inspiratory flow pattern were characterized by greater tidal volume (larger inspiratory flow-time area) and prolonged mechanical inflation time (*double edge blue arrows*). Inspection of the P_{eso} waveform does not provide sufficient information to identify the mechanism for the biphasic inspiratory flow pattern.

without electromyography, it is not possible to accurately quantify expiratory muscle contraction when it begins during inspiration, underestimating its value to an unknown extent.[9,28] We are confident that the observed ΔPgas was accurately distinguished from cardiac oscillations because we used predefined criteria, which excluded interference from cardiac impulses and because any apparent expiratory muscle contraction persisted for more than one-third of the respiratory cycle. Furthermore, we assessed respiratory drive by calculating both dPdi/dt and dP_{eso}/dt.[29] We refrained from assessing inspiratory muscle pressure due to the need for estimating chest wall elastance, which can be inherently unreliable when active expiration is present, often resulting in substantial deviations from predicted values or values measured during passive ventilation. There are no target values for dP_{eso}/dt and dPdi/dt in mechanically ventilated patients. Previous studies from our group have shown that under assisted ventilation dPdi/dt was higher than 7 cmH_2O/sec with a high correlation between the magnitude of inspiratory effort and dPdi/dt.[19,24] Based on these and former observations on dP_{eso}/dt,[25] and considering high effort as diaphragmatic pressure swing greater than 10 cmH_2O,[30] we selected a threshold greater than 10 cmH_2O/sec for both dPdi/dt and dP_{eso}/dt to define high respiratory drive.

Active Expiration and Respiratory Drive

The assessment of respiratory drive is complicated in critically ill patients. Patients with a high respiratory drive may have a low dPdi/dt and dP_{eso}/dt if there is impaired neuromuscular transmission of respiratory center output to the respiratory muscles. Furthermore, dPdi/dt may underestimate the respiratory drive more than dP_{eso}/dt because (a) it does not take into account the recruitment of accessory inspiratory and expiratory muscles and (b) the conversion of respiratory drive into diaphragmatic pressure is contingent upon the length of the diaphragm, its mechanical advantage, its strength and synergism of other muscles.[29,31,32] Indeed, dPdi/dt did not differ between conditions with active and passive expiration but dP_{eso}/dt, which incorporated the relaxation of expiratory muscles was significantly

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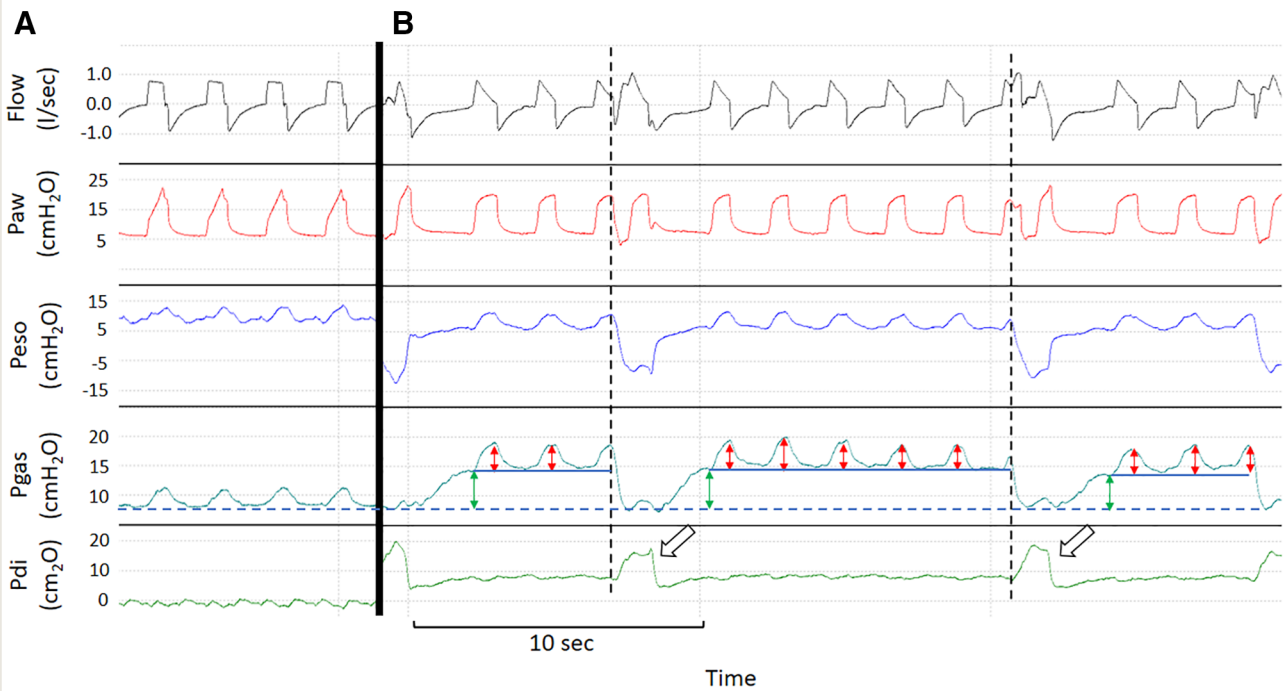

Fig. 7. Flow (l/min), airway pressure (P_{aw}), esophageal pressure (P_{eso}), gastric pressure (Pgas) and transdiaphragmatic pressure (Pdi) in a patient ventilated with volume control and muscle relaxants (*A*) and with bilevel positive pressure ventilation after withdrawing muscle relaxation (*B*; BIPAP). Waveforms indicate totally passive ventilation during volume control, while in BIPAP passive ventilation is interrupted by sporadic inspiratory efforts (*open arrows*). Inspection of Pgas waveform reveals that, during BIPAP, and between inspiratory efforts, expiratory muscles exhibit a sustained (tonic) contraction lasting more than 5 s. The phasic increase in Pgas above that achieved by tonic activity (*vertical double edge red arrows* in *B*) is mainly due to passive increase in volume since the phasic change in Pgas is closely similar to that observed with volume control (*vertical double edge blue arrows* in *A*). Notice that with BIPAP inspiratory efforts immediately follow relaxation of expiratory muscle activity (*vertical dashed black line*), which is indicated by the decrease in Pgas to level observed during volume control (*horizontal dashed blue line*). After diaphragm relaxation, Pgas increases rapidly from 8 to 15 cmH_2O (*vertical double edges green arrows*).

higher. It should be also noted that factors unrelated to respiratory drive may trigger the abdominal muscles. For example, several studies have shown that opioids and PEEP, a drug commonly used in mechanically ventilated patients, can stimulate expiratory muscles with increases in abdominal pressure of up to 20 cmH_2O.[33–37]

Clinical Significance of Identifying Abnormal Patient–Ventilator Interaction Associated With Expiratory Muscles

Numerous adverse consequences arise from abnormal patient–ventilator interaction, compelling clinicians to detect and correct it through a thorough analysis of flow and P_{aw} ventilator waveforms. Nevertheless, the focus is almost exclusively on the mismatch between inspiratory effort and pressure delivery while the role of the expiratory muscles is largely neglected. Failure to recognize active expiration as the cause of impaired patient–ventilator interaction could influence clinical management in the wrong direction.

The majority of patients with ineffective effort-like distortions had high respiratory drive. If these distortions were attributed to ineffective efforts, the clinician might choose to reduce the assist level, apply external PEEP, cycle off the ventilator earlier, or increase trigger sensitivity.[1,4,38,39] However, reducing the assist level could be inappropriate in the presence of high respiratory drive and expiratory muscle contraction. In addition, expiratory muscle recruitment could be further exacerbated by the escalation of external PEEP.[40–43] Finally, an increase in trigger sensitivity could promote or enhance triggering associated with partial expiratory muscle relaxation.

In some patients, expiratory gastric pressure fluctuations occurred long before expiratory muscle relaxation and triggered the ventilator, in the same way that leaks, cardiac oscillations, water in the circuit, or secretions cause autotriggering.[23,44] Flow-triggered autotriggering arises when flow increases from zero. Thus, when flow remains at zero for an extended period, the system becomes susceptible to autotriggering. Patients with low respiratory drive, often characterized by a low respiratory rate, are at increased risk of autotriggering due to prolonged expiratory time.[1,45] Our patients, had low respiratory rate and zero expiratory flow for some time before the subsequent

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Table 3. Expiratory Effort, Inspiratory Effort, Ventilation Parameters, and Respiratory Drive

Parameters	No. of Measurements	Value
ΔPgas, cmH_2O	90	3.6 (2.7–5.5)
ΔPdi, cmH_2O	118	8.4 (5.4–11.2)
ΔP_{eso}, cmH_2O	123	10.0 (7.4–13.7)
Duration of active expiration, s	90	1.69 (1.27–2.35)
V_T, ml	123	475 (419–556)
RR, br/min	123	20.5 (16.5–24.1)
$\dot{V}_E$, lt/min	123	9.9 (8.2–12.9)
MIP, cmH_2O	67	31 (22–47)
dPdi/dt, cm H_2O/s	118	9.8 (6.6–15.5)
dP_{eso}/dt, cm H_2O/s	123	11.6 (8.1–17.4)
P0.1, cmH_2O	63	3.2 (1.8–4.1)

Values are median (1st, 3rd quartile).

dPdi/dt, rate of diaphragmatic pressure increase; dP_{eso}/dt, rate of esophageal pressure decrease; MIP, maximum inspiratory pressure; P 0.1, occlusion pressure at 100 msec; ΔPdi, inspiratory increase in diaphragmatic pressure; ΔP_{eso}, inspiratory decrease in esophageal pressure; ΔPgas, expiratory increase in gastric pressure; RR, respiratory rate; $\dot{V}_E$, minute ventilation; V_T, tidal volume.

Table 4. Comparison between Active Expiration and Passive Expiration

No. of Conditions	Active Expiration	Passive Expiration
ΔPdi, cmH_2O	8.03 (5.4–11.7)	9.3 (4.9–11.6)
ΔP_{eso}, cmH_2O	10.7 (8.2–14.6)	8.4 (5.8–12.0)*
V_T, ml	468 (417–545)	497 (443–615)
RR, br/min	22 (17–25)	19 (17–23)
PEEP, cmH_2O	8.0 (7.0–8.0)	7.5 (6.0–9.0)
Assist level, cmH_2O	5 (0–8)	5 (5–8)
dPdi/dt, cmH_2O/s	9.3 (6.2–15.8)	9.9 (7.0–15.7)
dP_{eso}/dt, cmH_2O/s	12.7 (9.0–18.5)	9.2 (6.8–14.2)*
P0.1, cmH_2O	3.4 (1.8–4.2)	2.1 (1.6–3.5)
MIP, cmH_2O	31 (21.8–50.0)	32 (23.5–40.0)
Fentanyl dose, mcg/h	150 (0–300)	100 (0–250)

Values are median (1st, 3rd quartile).

*$P < 0.05$ between the two groups.

dPdi/dt, rate of diaphragmatic pressure increase; dP_{eso}/dt, rate of esophageal pressure decrease; MIP, maximum inspiratory pressure; P 0.1, occlusion pressure at 100 msec; PEEP, positive end-expiratory pressure; ΔPdi, inspiratory increase in diaphragmatic pressure; ΔP_{eso}, inspiratory decrease in esophageal pressure; ΔPgas, expiratory increase in gastric pressure; RR, respiratory rate; V_T, tidal volume.

mechanical inflation. At the same time, the inspiratory flow-time curve indicated a lack of inspiratory effort during pressure delivery (figs. 3 and 4C). Therefore, there was no distinguishing feature between autotriggering and triggering by gastric pressure fluctuation.[38] Moreover, if autotriggering along with low respiratory rate were considered as indications of low respiratory drive, a possible intervention would be the reduction of the level of support. However, half of the patients with gastric pressure-induced triggering had high respiratory drive and reduction of ventilator assist could, in theory exaggerate the contraction of the expiratory muscles.

Moreover, expiratory gastric pressure fluctuations during active expiration caused multiple triggering in 10 patients. Multiple triggering is usually addressed by adjusting the cycling-off criterion or increasing ventilatory support.[1,4,39] Higher levels of support in multiple triggering associated with expiratory muscle relaxation can either mitigate multiple triggering, particularly when active expiration stems from insufficient assistance, or potentially exacerbate V_T and minute ventilation without ameliorating active expiration.[46]

Recognition of Expiratory Muscle Recruitment

Passive expiration is evidenced by a peak early in expiration followed by an exponential decay of the expiratory flow-time curve toward zero (in the absence of dynamic hyperinflation). The peak expiratory flow and the slope of the decay are determined by the mechanical properties of the respiratory system and endotracheal tube.[47] In the presence of expiratory muscle activity, the peak expiratory flow appears "decapitated" with a more rounded shape, the post-peak expiratory flow is higher and may not reach zero before the next inspiration. However, as seen in several patients with expiratory muscle contraction, the shape of the expiratory flow may not differ from that of passive expiration, and the expiratory flow may reach zero in the presence of airway closure or as the end-expiratory lung volume approaches residual capacity. Therefore, observation of ventilator waveforms may not always reveal active expiration.

Limitations and Open Questions

The single-center, retrospective nature of the study is inherent to several limitations. It assessed patients who required P_{eso} or diaphragmatic pressure monitoring for clinical purposes. Hence, the presence of active expiration could be higher than the general ICU population. Although the interpretation of complicated waveforms might be subjective at times, we are confident that this limitation is not a significant issue. All waveforms were meticulously examined by four investigators with extensive experience in waveform analysis and interpretation. Notably, there was no disagreement among them in their observations. As already discussed, our study cannot suggest if and how abnormal patient–ventilator interaction related to expiratory muscle activity should be managed. Furthermore, factors associated with active expiration were retrospectively collected and available data are not sufficient to make conclusions of their relationship with active expiration. A systematic investigation of other parameters such as hypercapnia, elastic and resistive load of the respiratory system, medications other than opioids, pain, delirium, and behavioral or other stimuli that might induce active expiration was not available. Finally, the effects of expiratory muscle contraction on the lungs, diaphragm, and cardiovascular status, as well as the impact of

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active expiration on weaning and ICU outcomes, require further investigation.

Conclusions

In mechanically ventilated patients with spontaneous breathing activity, monitored with P_{eso} and diaphragmatic pressure, expiratory muscle contraction was frequent and was associated with events of abnormal patient–ventilator interaction and high respiratory drive. Even with advanced monitoring using P_{eso}, it may be impossible to recognize events associated with active expiration. Without gastric pressure monitoring, they can be erroneously misinterpreted as a mismatch between inspiratory effort and mechanical inflation. However, it is crucial to recognize that the underlying mechanism and appropriate interventions for these events may be different.

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Competing Interest

Dr. Akoumianaki, Dr. Georgopoulos, and Dr. Vaporidi have received travel and speaker fees from Medtronic (Dubai, United Arab Emirates). The other authors declare no competing interests.

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Supplemental Digital Content

Supplemental Digital Content: https://links.lww.com/ALN/D570
1. Supplemental Methods and Results
2. Supplemental Tables E1-E3
3. Supplemental Figure E1
4. Supplemental Figure E2

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